

Scholaro's GCSE Biology Guide: Topic 4 - Bioenergetics (Made for ESL Students)

Welcome to Your Bioenergetics ESL Guide!

Purpose of This Guide

This guide explains **Bioenergetics**, the fourth topic in GCSE AQA Biology. You'll learn about **photosynthesis**, **respiration**, how **exercise affects the body**, and how cells use energy. Everything is explained step-by-step with easy examples.

What You Will Learn

-  Photosynthesis and its factors
 -  How plants use glucose
 -  Respiration (aerobic and anaerobic)
 -  Exercise and the body
 -  Metabolism
-

1. Photosynthesis

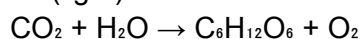
Photosynthesis is the process where **plants make glucose** using **sunlight**.

Type of Reaction:

- It is **endothermic** (takes in energy)
- Happens in **chloroplasts** using light energy

Equation:

Carbon dioxide + Water → Glucose + Oxygen
(light)





2. Rate of Photosynthesis



Factors That Affect It:

Factor	What it Does
Light intensity	More light = faster photosynthesis
Temperature	Warmer = faster (until too hot, then enzymes break)
Carbon dioxide	Needed to make glucose
Chlorophyll	The green pigment that absorbs light



Practical: Oxygen Bubble Test

- Put pondweed in water with a light source
- Measure bubbles = oxygen made
- Change 1 factor (light, temp, etc.) at a time



Limiting Factors:

- If one factor is too low, it **limits** the rate
- Graphs show this when lines level off



Inverse Square Law:

Light intensity $\propto 1 / \text{distance}^2$

If you move the lamp **twice** as far, light = $\frac{1}{4}$ as strong



3. How Plants Use Glucose

Use	Purpose
Respiration	Energy for plant processes
Stored as starch	For use when no light (night)
Make fats/oils	Stored in seeds
Make cellulose	To build strong cell walls
Make proteins	With nitrates from soil

❤️ 4. Respiration

Respiration happens in **all cells** to release **energy** from glucose.

✨ Type of Reaction:

- It is **exothermic** (releases energy)

🌀 Types of Respiration:

Type	With Oxygen?	Equation	Energy?
Aerobic	Yes	Glucose + O ₂ → CO ₂ + H ₂ O	High
Anaerobic (Animals)	No	Glucose → Lactic acid	Low
Anaerobic (Yeast)	No	Glucose → Ethanol + CO ₂	Low

🪑 Used in:

- **Bread and beer** making (yeast fermentation)
-

🏃 5. Exercise and the Body

During exercise, muscles need more **energy**, so:

- **Heart rate** increases
- **Breathing rate** increases
- **Breath volume** increases

If not enough oxygen gets to muscles:

- **Anaerobic respiration** starts
- **Lactic acid** builds up
- This causes **muscle fatigue**

😓 Oxygen Debt:

Extra oxygen needed **after** exercise to:

- Break down lactic acid
- Return body to normal

Transport:

- Lactic acid goes in blood to **liver**, turned into glucose
-



6. Metabolism

Metabolism is **all the chemical reactions** in your body.

- Energy from respiration is used in metabolism



Examples:

Reaction	What it Does
Glucose → Starch/cellulose/glycogen	Storage/structure
Glycerol + fatty acids → Lipids	Makes fat molecules
Glucose + nitrate → Amino acids	Build proteins
Respiration	Releases energy
Breakdown of proteins → Urea	For excretion



Summary

You learned:

- How plants make food using sunlight
 - What affects the speed of photosynthesis
 - How glucose is used
 - Differences between aerobic and anaerobic respiration
 - What happens during exercise
 - What metabolism means
-



Amazing job! Bioenergetics helps you understand where energy comes from in living things.

Made with ❤️ by Scholaro for ESL students studying AQA GCSE Biology.